

EC-231 Operating Systems

BS(CE)-2021

LAB REPORT # 4



Submitted By:

Reg. No.	Name
21-CE-009	Huzaifa Shahbaz

Department of Computer Engineering

BS Computer Engineering Program

HITEC University Taxila

	Presentation (Format (0.5) + Title (0.5) + Conclusion(1) + Procedure(2) + Objective(1))	Calculation/ Coding	Observation/ Program Results	Total
Total	5	5	5	15
Obtained				

Lab Instructor.
Engr. Fasih Ahmad

Experiment #4

Implementation of Conditional Statements and Loops in Linux

Objective:

The objectives of this lab are to familiarize students with shell programming and practice few problems related to

- 1: If-else statements
 - 2: For loop
 - 3: While loop
- in LINUX Ubuntu.

Software Used:

1. Ubuntu
2. gedit/vi(Text Editor)

Procedure:

This lab has been designed to get even better insight of the shell programming. By the end of this lab we'll be able to solve more complex problems in Linux using shell. The details of few programming constructs to be used in the tasks are as follows:

Mathematical Operator in Shell Script	Meaning	Normal Arithmetical / Mathematical Statements	But in Shell	
			For test statement with if command	For [expr] statement with if command
-eq	Is equal to	$a==b$	if test \$a -eq \$b if test \$a -eq \$b;	if [[\$a -eq \$b]] if [\$a -eq \$b];
-ne	Not equal to	$a!=b$	if test \$a -ne \$b if test \$a -ne \$b;	if [[\$a -ne \$b]] if [\$a -ne \$b];
-lt	Less than	$a<b$	if test \$a -lt \$b if test \$a -lt \$b;	if [[\$a -lt \$b]] if [\$a -lt \$b];
-le	Less than or equal to	$a<=b$	if test \$a -le \$b if test \$a -le \$b;	if [[\$a -le \$b]] if [\$a -le \$b];
-gt	Greater than	$a>b$	if test \$a -gt \$b if test \$a -gt \$b;	if [[\$a -gt \$b]] if [\$a -gt \$b];
-ge	Greater than or equal to	$a>=b$	if test \$a -ge \$b if test \$a -ge \$b;	if [[\$a -ge \$b]] if [\$a -ge \$b];

if statements :

```
if condition then
commands
fi
```

Example:

Following script determine whether given argument number is positive.

```
if test $1 -gt 0 then
```

```
echo "$1 number is positive" fi
```

Run it as follows:

```
$ chmod 755  
ispostive $./ ispostive  
5
```

```
5 number is positive
```

```
$/ispostive -45
```

```
Nothing is printed
```

if else statements:

```
ifcondition then
```

```
commands
```

```
else
```

```
commands
```

```
fi
```

Example:

Following script determine whether given argument number is positive.

```
$ cat>ispostive  
#  
#  
# Script to see whether argument is positive  
#  
number=10  
if [ $number -eq 10 ]; then  
echo "Number is 10."  
Else  
echo "Number is not 10."  
Fi
```

Run it as follows:

```
$ chmod 755  
ispostive $./ ispostive  
5  
Number is 10
```

Case...esac statement in shell:

These are the statements in shell which replicates the switch case statements in c/c++ language.

Syntax is as follows:

```
#!/bin/bash  
# Example variable  
option="B"  
case "$option" in  
"A")  
echo "Option A selected"  
# Add commands to execute when option A is selected
```

```
;;
"B")
echo "Option B selected"
# Add commands to execute when option B is selected
;;
"C")
echo "Option C selected"
# Add commands to execute when option C is selected
;;
*)
echo "Invalid option"
# Add commands to execute when none of the specified options match
;;
esac
```

While loop:

```
while [ condition ]
do
command1
command2
..
....
commandN
done
```

Example:

```
count=1
# While loop
while [ $count -le 5 ]; do
echo "Count: $count"
count=$((count + 1))
done
```

Tasks

Task 1:

Simulate suitable command/commands in gedit to input 2 numbers from user and display the greater number.

CODE:

```
1 #!/bin/bash
2 echo "Calculator"
3 echo "Enter the first number:"
4 read num1
5 if [ "$num1" -gt 0 ];
6 then
7 echo "It is a positive number"
8 elif [ $num1 -ge 0 ];
9 then
10 echo "It is 0"
11 else
12 echo "It is a Negative Number"
13 fi
```

OUTPUT:

```
huzaifashahbaz@huzaifashahbaz-VirtualBox:~$ pwd
/home/huzaifashahbaz
huzaifashahbaz@huzaifashahbaz-VirtualBox:~$ ls
Android      Desktop      Downloads    Music  Pictures  snap      Videos
AndroidStudioProjects  Documents    file6.sh   name    Public    Templates
huzaifashahbaz@huzaifashahbaz-VirtualBox:~$ mkdir huzaifa702
huzaifashahbaz@huzaifashahbaz-VirtualBox:~$ cd huzaifa702
huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa702$ touchfile1.sh
touchfile1.sh: command not found
huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa702$ touch file1.sh
huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa702$ ls -l
total 0
-rw-rw-r-- 1 huzaifashahbaz huzaifashahbaz 0 16:20 23  مارچ file1.sh
huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa702$ chmod 777 file1.sh
huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa702$ xdg-open file1.sh
```

```
Enter the first number:
-5
```

```
It is a negative number
```

Task 2:

Simulate suitable command/commands in gedit to input 2 numbers from user and make a simple calculator which should be capable of performing basic calculations i.e. addition, subtraction, multiplication, division by using if-else conditions. Display all possible outputs.

CODE:

```

1 #!/bin/bash
2 echo "Enter the first number:"
3 read num1
4 echo "Enter the second number:"
5 read num2
6 echo "Select operation:"
7 echo "1.Addition"
8 echo "2.Subtraction"
9 echo "3.Multiplication"
10 echo "4.Division:"
11 read choice
12 if [$choice -eq 1];then
13     result=$(echo "$num1 +$num2" |bc)
14     echo "Result:$result"
15 elif [$choice -eq 2];then
16     result=$(echo "$num1 -$num2" |bc)
17     echo "Result:$result"
18 elif [$choice -eq 3];then
19     result=$(echo "$num1 *$num2" |bc)
20     echo "Result:$result"
21 elif [$choice -eq 4];then
22     result=$(echo "scale=2; $num1/$num2" |bc)
23     echo "Result:$result"
24 else
25     echo "Invalid choice"
26 fi

```

OUTPUT:

```

huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa702$ touch file2.sh
huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa702$ ls -l
total 4
-rwxrwxrwx 1 huzaifashahbaz huzaifashahbaz 167 16:35 23  ﻣﺎﺝ file1.sh
-rw-rw-r-- 1 huzaifashahbaz huzaifashahbaz  0 16:38 23  ﻣﺎﺝ file2.sh
huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa702$ chmod 777 file2.sh
huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa702$ xdg-open file2.sh
huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa702$ ./file2.sh

```

```

Enter first number:
5
Enter second number:
5
Select operation:
1. Addition
2. Subtraction
3. Multiplication
4. Division
1
Result: 10

```

Task 3:

Simulate suitable command/commands in gedit to input 2 numbers from user and make a simple calculator which should be capable of performing basic calculations i.e. addition,

subtraction, multiplication, division by using switch-case statements. Display all possible outputs.

CODE:

```
1 #!/bin/bash
2 echo "Enter the first number:"
3 read num1
4 echo "Enter the second number:"
5 read num2
6 echo "Select operation"
7 echo "1.Addition"
8 echo "2.Subtraction"
9 echo "3.Multiplication"
10 echo "4.Division"
11 read choice
12 case $choice in
13     1)result=$(echo "$num1 + $num2" | bc)
14     ;;
15     2)result=$(echo "$num1 - $num2" | bc)
16     ;;
17     3)result=$(echo "$num1 * $num2" | bc)
18     ;;
19     4)result=$(echo "scale=2; $num1/$num2" | bc)
20     ;;
21     *)echo "Invalid choice"
22     exit 1
23     ;;
24     esac
25     echo "Results:$result"
26 fi
```

OUTPUT:

```
huzaihashahbaz@huzaihashahbaz-VirtualBox:~/huzai702$ touch file3.sh
huzaihashahbaz@huzaihashahbaz-VirtualBox:~/huzai702$ ls -l
total 8
-rwxrwxrwx 1 huzaihashahbaz huzaihashahbaz 167 16:35 23   ﻣﺎﺭﭼ  file1.sh
-rwxrwxrwx 1 huzaihashahbaz huzaihashahbaz 760 16:55 23   ﻣﺎﺭﭼ  file2.sh
-rw-rw-r-- 1 huzaihashahbaz huzaihashahbaz   0 16:58 23   ﻣﺎﺭﭼ  file3.sh
huzaihashahbaz@huzaihashahbaz-VirtualBox:~/huzai702$ chmod 777 file3.sh
huzaihashahbaz@huzaihashahbaz-VirtualBox:~/huzai702$ xdg-open file3.sh
huzaihashahbaz@huzaihashahbaz-VirtualBox:~/huzai702$ ./file3.sh
```

```
Enter the first number:
7
Enter the second number:
7
Select operation
1.Addition
2.Subtraction
3.Multiplication
4.Division
1
Results:14
```

```
Enter the first number:
7
Enter the second number:
7
Select operation
1.Addition
2.Subtraction
3.Multiplication
4.Division
2
Results:0
```

```
Enter the first number:
7
Enter the second number:
7
Select operation
1.Addition
2.Subtraction
3.Multiplication
4.Division
3
Results:49
```

```
Enter the first number:
7
Enter the second number:
7
Select operation
1.Addition
2.Subtraction
3.Multiplication
4.Division
4
Results:1.00
```

```
Enter the first number:
7
Enter the second number:
7
Select operation
1.Addition
2.Subtraction
3.Multiplication
4.Division
*
Invalid choice
```

Task 4:

Simulate suitable command/commands in gedit to print table of a user defined number.

CODE:

```
1 #!/bin/bash
2 # Prompt the user to enter a number
3 read -p "Enter a number to generate a table: " num
4 # Print the table header
5 echo "Table of $num:"
6 # Generate and print the table
7 for (( i=1; i<=10; i++ ))
8 do
9     echo "$num * $i = $((num * i))"
10 done
```

OUTPUT:

```
huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa702$ touch file4.sh
huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa702$ ls -l
total 12
-rwxrwxrwx 1 huzaifashahbaz huzaifashahbaz 167 16:35 23   ماج file1.sh
-rwxrwxrwx 1 huzaifashahbaz huzaifashahbaz 760 16:55 23   ماج file2.sh
-rwxrwxrwx 1 huzaifashahbaz huzaifashahbaz 528 17:16 23   ماج file3.sh
-rw-rw-r-- 1 huzaifashahbaz huzaifashahbaz   0 17:18 23   ماج file4.sh
huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa702$ chmod 777 file4.sh
huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa702$ xdg-open file4.sh
huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa702$ ./file4.sh
```

```
Enter a number to generate a table: 4
Table of 4:
4 * 1 = 4
4 * 2 = 8
4 * 3 = 12
4 * 4 = 16
4 * 5 = 20
4 * 6 = 24
4 * 7 = 28
4 * 8 = 32
4 * 9 = 36
4 * 10 = 40
```

Task 5:

Simulate suitable command/commands in gedit to check whether the user defined number is even or odd.

CODE:

```
1 #!/bin/bash
2 echo "Enter a number:"
3 read number
4 if [ $((number % 2)) -eq 0 ]; then
5     echo "$number is even"
6 else
7     echo "$number is odd"
8 fi
```

OUTPUT:

```
huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa702$ touch file5.sh
huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa702$ ls -l
total 16
-rwxrwxrwx 1 huzaifashahbaz huzaifashahbaz 167 16:35 23   ماج file1.sh
-rwxrwxrwx 1 huzaifashahbaz huzaifashahbaz 760 16:55 23   ماج file2.sh
-rwxrwxrwx 1 huzaifashahbaz huzaifashahbaz 528 17:16 23   ماج file3.sh
-rwxrwxrwx 1 huzaifashahbaz huzaifashahbaz 247 17:21 23   ماج file4.sh
-rw-rw-r-- 1 huzaifashahbaz huzaifashahbaz   0 17:36 23   ماج file5.sh
huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa702$ chmod 777 file5.sh
huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa702$ xdg-open file5.sh
huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa702$ ./file5.sh
Enter a number:
8
8 is even
```

Conclusion:

In conclusion, we had studied the conditions that is if statement, if-else statement, else-if statement and else statement. And we had studied the switch statement and also for loop and while loop. There functions and working in ubuntu operating system.